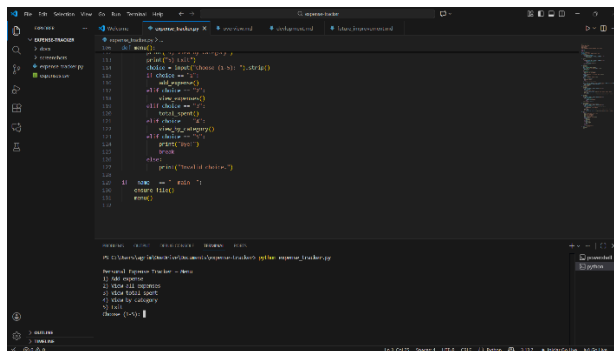


SCREENSHOTS

To show the running of the program and working of the terminal -

1). Screenshot of the Running of the program in the terminal through Gitbash -



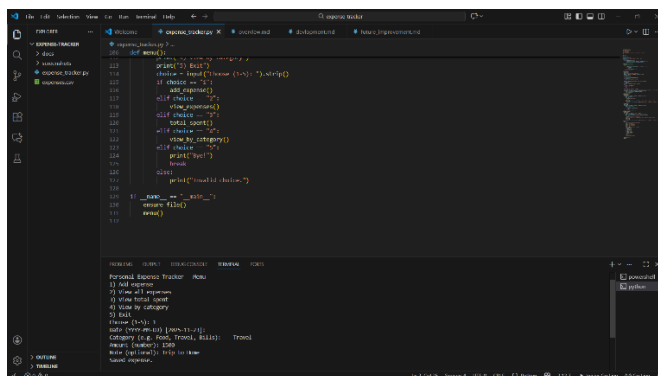
The screenshot shows a code editor with a dark theme. The main window displays the source code of a program. The code is in C++ and includes a menu-driven interface for a personal expense tracker. The code is as follows:

```
1 // Personal Expense Tracker
2 #include <iostream>
3 #include <string>
4 #include <vector>
5 #include <map>
6 #include <algorithm>
7 #include <ctime>
8 #include <cstdlib>
9 #include <conio.h>
10 using namespace std;
11
12 struct Expense {
13     string date;
14     string category;
15     double amount;
16 };
17
18 vector<Expense> expenses;
19
20 void displayMenu() {
21     cout << "Personal Expense Tracker - Menu\n";
22     cout << "1. Add Expense\n";
23     cout << "2. View All Expenses\n";
24     cout << "3. View Total Spent\n";
25     cout << "4. View by Category\n";
26     cout << "5. Exit\n";
27 }
28
29 void addExpense() {
30     Expense e;
31     cout << "Enter date (dd-mm-yyyy): ";
32     string date;
33     while (date.empty() || date.length() > 10) {
34         cout << "Invalid date format. Try again: ";
35         date = "";
36     }
37     e.date = date;
38     cout << "Enter category: ";
39     string category;
40     while (category.empty() || category.length() > 50) {
41         cout << "Invalid category. Try again: ";
42         category = "";
43     }
44     e.category = category;
45     cout << "Enter amount: ";
46     double amount;
47     while (amount <= 0) {
48         cout << "Invalid amount. Try again: ";
49         amount = 0;
50     }
51     e.amount = amount;
52     expenses.push_back(e);
53     cout << "Expense added successfully.\n";
54 }
55
56 void viewAllExpenses() {
57     if (expenses.empty()) {
58         cout << "No expenses found.\n";
59     } else {
60         cout << "All Expenses:\n";
61         for (const auto& e : expenses) {
62             cout << e.date << " | " << e.category << " | " << e.amount << "\n";
63         }
64     }
65 }
66
67 void viewTotalSpent() {
68     double total = 0;
69     for (const auto& e : expenses) {
70         total += e.amount;
71     }
72     cout << "Total Spent: " << total << "\n";
73 }
74
75 void viewByCategory() {
76     if (expenses.empty()) {
77         cout << "No expenses found.\n";
78     } else {
79         cout << "View by Category:\n";
80         map<string, double> categoryTotal;
81         for (const auto& e : expenses) {
82             categoryTotal[e.category] += e.amount;
83         }
84         for (const auto& p : categoryTotal) {
85             cout << p.first << " : " << p.second << "\n";
86         }
87     }
88 }
89
90 int main() {
91     int choice;
92     do {
93         displayMenu();
94         choice = 0;
95         while (choice < 1 || choice > 5) {
96             cout << "Invalid choice. Try again: ";
97             choice = 0;
98         }
99         switch (choice) {
100             case 1: addExpense(); break;
101             case 2: viewAllExpenses(); break;
102             case 3: viewTotalSpent(); break;
103             case 4: viewByCategory(); break;
104             case 5: exit(0);
105         }
106     } while (choice != 5);
107     return 0;
108 }
```

The terminal output shows the program running and the user interacting with the menu. The output is as follows:

```
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 1
Enter date (dd-mm-yyyy): 10-10-2023
Enter category: Travel
Enter amount: 1000
Expense added successfully.
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 2
All Expenses:
10-10-2023 | Travel | 1000
Enter choice: 3
Total Spent: 1000
Enter choice: 4
View by Category:
Travel : 1000
Enter choice: 5
Exit
```

2). First output of the program



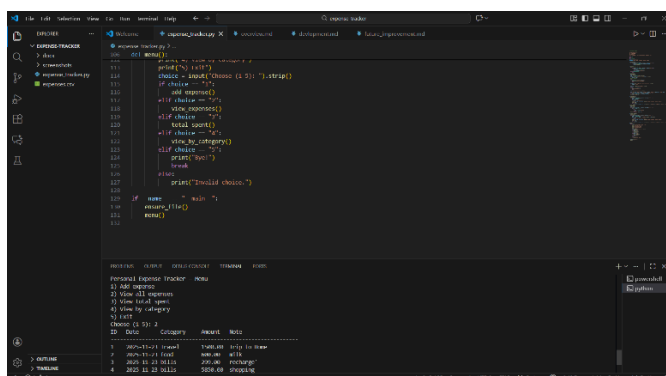
The screenshot shows a code editor with a dark theme. The main window displays the source code of a program. The code is in C++ and includes a menu-driven interface for a personal expense tracker. The code is as follows:

```
1 // Personal Expense Tracker
2 #include <iostream>
3 #include <string>
4 #include <vector>
5 #include <map>
6 #include <algorithm>
7 #include <ctime>
8 #include <cstdlib>
9 #include <conio.h>
10 using namespace std;
11
12 struct Expense {
13     string date;
14     string category;
15     double amount;
16 };
17
18 vector<Expense> expenses;
19
20 void displayMenu() {
21     cout << "Personal Expense Tracker - Menu\n";
22     cout << "1. Add Expense\n";
23     cout << "2. View All Expenses\n";
24     cout << "3. View Total Spent\n";
25     cout << "4. View by Category\n";
26     cout << "5. Exit\n";
27 }
28
29 void addExpense() {
30     Expense e;
31     cout << "Enter date (dd-mm-yyyy): ";
32     string date;
33     while (date.empty() || date.length() > 10) {
34         cout << "Invalid date format. Try again: ";
35         date = "";
36     }
37     e.date = date;
38     cout << "Enter category: ";
39     string category;
40     while (category.empty() || category.length() > 50) {
41         cout << "Invalid category. Try again: ";
42         category = "";
43     }
44     e.category = category;
45     cout << "Enter amount: ";
46     double amount;
47     while (amount <= 0) {
48         cout << "Invalid amount. Try again: ";
49         amount = 0;
50     }
51     e.amount = amount;
52     expenses.push_back(e);
53     cout << "Expense added successfully.\n";
54 }
55
56 void viewAllExpenses() {
57     if (expenses.empty()) {
58         cout << "No expenses found.\n";
59     } else {
60         cout << "All Expenses:\n";
61         for (const auto& e : expenses) {
62             cout << e.date << " | " << e.category << " | " << e.amount << "\n";
63         }
64     }
65 }
66
67 void viewTotalSpent() {
68     double total = 0;
69     for (const auto& e : expenses) {
70         total += e.amount;
71     }
72     cout << "Total Spent: " << total << "\n";
73 }
74
75 void viewByCategory() {
76     if (expenses.empty()) {
77         cout << "No expenses found.\n";
78     } else {
79         cout << "View by Category:\n";
80         map<string, double> categoryTotal;
81         for (const auto& e : expenses) {
82             categoryTotal[e.category] += e.amount;
83         }
84         for (const auto& p : categoryTotal) {
85             cout << p.first << " : " << p.second << "\n";
86         }
87     }
88 }
89
90 int main() {
91     int choice;
92     do {
93         displayMenu();
94         choice = 0;
95         while (choice < 1 || choice > 5) {
96             cout << "Invalid choice. Try again: ";
97             choice = 0;
98         }
99         switch (choice) {
100             case 1: addExpense(); break;
101             case 2: viewAllExpenses(); break;
102             case 3: viewTotalSpent(); break;
103             case 4: viewByCategory(); break;
104             case 5: exit(0);
105         }
106     } while (choice != 5);
107     return 0;
108 }
```

The terminal output shows the program running and the user interacting with the menu. The output is as follows:

```
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 1
Enter date (dd-mm-yyyy): 10-10-2023
Enter category: Travel
Enter amount: 1000
Expense added successfully.
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 2
All Expenses:
10-10-2023 | Travel | 1000
Enter choice: 3
Total Spent: 1000
Enter choice: 4
View by Category:
Travel : 1000
Enter choice: 5
Exit
```

3). Second output of the program



The screenshot shows a code editor with a dark theme. The main window displays the source code of a program. The code is in C++ and includes a menu-driven interface for a personal expense tracker. The code is as follows:

```
1 // Personal Expense Tracker
2 #include <iostream>
3 #include <string>
4 #include <vector>
5 #include <map>
6 #include <algorithm>
7 #include <ctime>
8 #include <cstdlib>
9 #include <conio.h>
10 using namespace std;
11
12 struct Expense {
13     string date;
14     string category;
15     double amount;
16 };
17
18 vector<Expense> expenses;
19
20 void displayMenu() {
21     cout << "Personal Expense Tracker - Menu\n";
22     cout << "1. Add Expense\n";
23     cout << "2. View All Expenses\n";
24     cout << "3. View Total Spent\n";
25     cout << "4. View by Category\n";
26     cout << "5. Exit\n";
27 }
28
29 void addExpense() {
30     Expense e;
31     cout << "Enter date (dd-mm-yyyy): ";
32     string date;
33     while (date.empty() || date.length() > 10) {
34         cout << "Invalid date format. Try again: ";
35         date = "";
36     }
37     e.date = date;
38     cout << "Enter category: ";
39     string category;
40     while (category.empty() || category.length() > 50) {
41         cout << "Invalid category. Try again: ";
42         category = "";
43     }
44     e.category = category;
45     cout << "Enter amount: ";
46     double amount;
47     while (amount <= 0) {
48         cout << "Invalid amount. Try again: ";
49         amount = 0;
50     }
51     e.amount = amount;
52     expenses.push_back(e);
53     cout << "Expense added successfully.\n";
54 }
55
56 void viewAllExpenses() {
57     if (expenses.empty()) {
58         cout << "No expenses found.\n";
59     } else {
60         cout << "All Expenses:\n";
61         for (const auto& e : expenses) {
62             cout << e.date << " | " << e.category << " | " << e.amount << "\n";
63         }
64     }
65 }
66
67 void viewTotalSpent() {
68     double total = 0;
69     for (const auto& e : expenses) {
70         total += e.amount;
71     }
72     cout << "Total Spent: " << total << "\n";
73 }
74
75 void viewByCategory() {
76     if (expenses.empty()) {
77         cout << "No expenses found.\n";
78     } else {
79         cout << "View by Category:\n";
80         map<string, double> categoryTotal;
81         for (const auto& e : expenses) {
82             categoryTotal[e.category] += e.amount;
83         }
84         for (const auto& p : categoryTotal) {
85             cout << p.first << " : " << p.second << "\n";
86         }
87     }
88 }
89
90 int main() {
91     int choice;
92     do {
93         displayMenu();
94         choice = 0;
95         while (choice < 1 || choice > 5) {
96             cout << "Invalid choice. Try again: ";
97             choice = 0;
98         }
99         switch (choice) {
100             case 1: addExpense(); break;
101             case 2: viewAllExpenses(); break;
102             case 3: viewTotalSpent(); break;
103             case 4: viewByCategory(); break;
104             case 5: exit(0);
105         }
106     } while (choice != 5);
107     return 0;
108 }
```

The terminal output shows the program running and the user interacting with the menu. The output is as follows:

```
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 1
Enter date (dd-mm-yyyy): 10-10-2023
Enter category: Travel
Enter amount: 1000
Expense added successfully.
Personal Expense Tracker - Menu
1. Add Expense
2. View All Expenses
3. View Total Spent
4. View by Category
5. Exit
Enter choice: 2
All Expenses:
10-10-2023 | Travel | 1000
Enter choice: 3
Total Spent: 1000
Enter choice: 4
View by Category:
Travel : 1000
Enter choice: 5
Exit
```

4). Third output of the program

```
expense_tracker.py > .
117 def menu():
118     print("\nMenu")
119     choice = input("Choose (1-5): ").strip()
120     if choice == "1":
121         add_expense()
122     elif choice == "2":
123         view_expenses()
124     elif choice == "3":
125         total_spent()
126     elif choice == "4":
127         view_by_category()
128     elif choice == "5":
129         print("bye!")
130         break
131     else:
132         print("Invalid choice.")
133
134 if __name__ == "__main__":
135     ensure_file()
136     menu()
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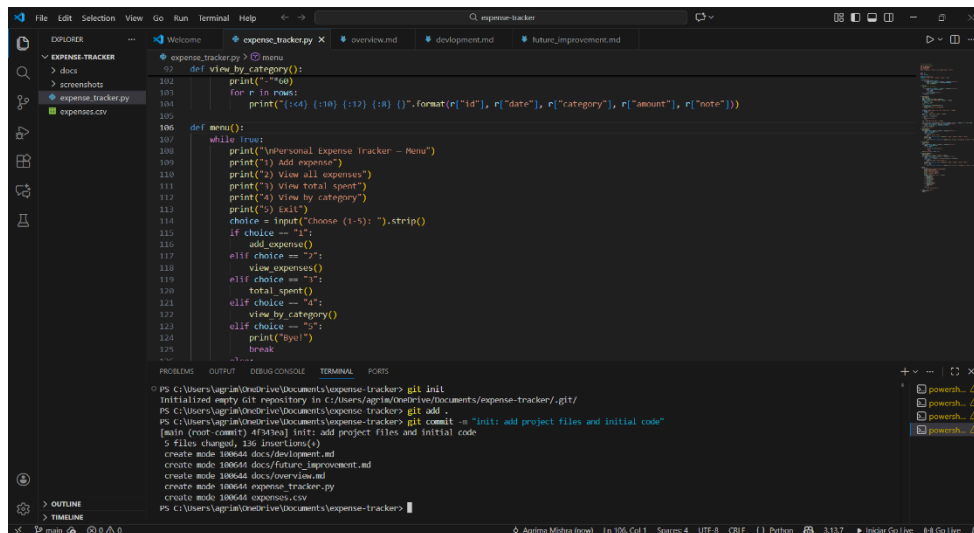
5). Fourth output of the program

```
expense_tracker.py > .
117 def menu():
118     print("\nMenu")
119     choice = input("Choose (1-5): ").strip()
120     if choice == "1":
121         add_expense()
122     elif choice == "2":
123         view_expenses()
124     elif choice == "3":
125         total_spent()
126     elif choice == "4":
127         view_by_category()
128     elif choice == "5":
129         print("bye!")
130         break
131     else:
132         print("Invalid choice.")
133
134 if __name__ == "__main__":
135     ensure_file()
136     menu()
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```

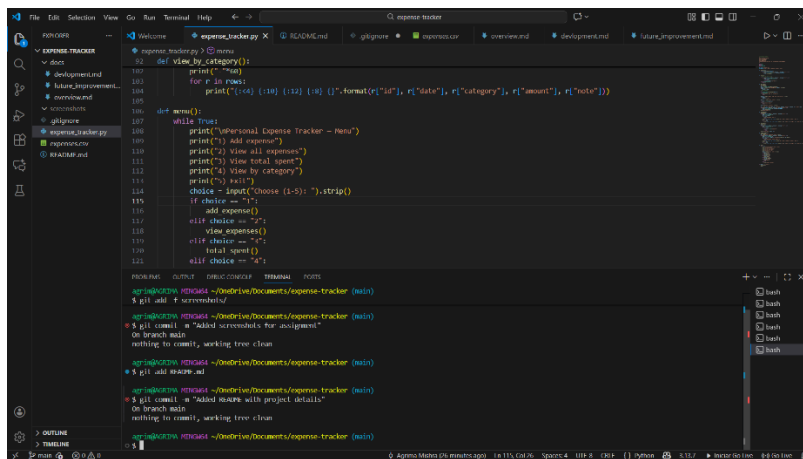
Commits –

1). git init

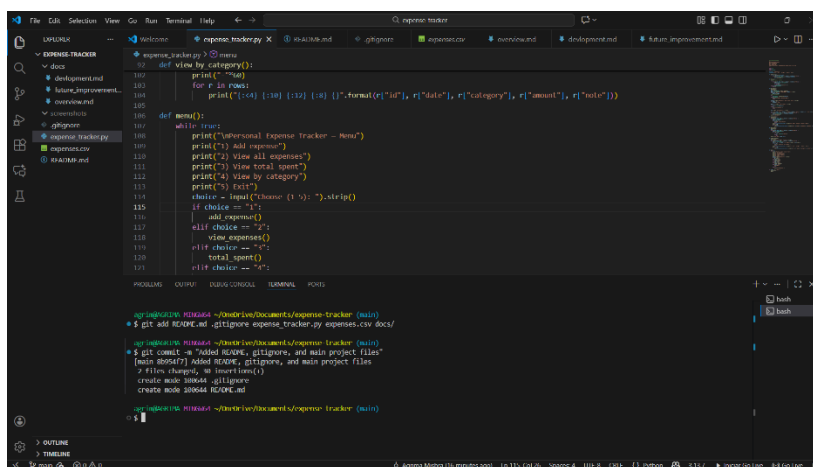
```
git init
Initialized existing Git repository in C:/Users/agria/OneDrive/Documents/expense-tracker/.git/
git init
Initialized existing Git repository in C:/Users/agria/OneDrive/Documents/expense-tracker/.git/
git add .
git commit -m "Initial commit: created project"
```



2). Git add screenshot



3). Git add README.md



```

C:\Users\user> git checkout -b main
Switched to a new branch 'main'

C:\Users\user> git commit -m "Initial commit"
[master (root-commit) 1234567] Initial commit
 2 files changed, 100 insertions(+), 0 deletions(-)
 create mode 100644 expense_tracker.py
 create mode 100644 README.md

C:\Users\user> git push -u origin main
Enumerating objects: 2, done.
Counting objects: 100%, 2/2 objects completed.
Compressing objects: 100%, 2/2 objects completed.
Writing objects: 100%, 2/2 objects completed.
Total 2 (delta 0), reused 0 (delta 0), packed 0 (delta 0), unpacked 2 (delta 0)
To https://github.com:username/expense-tracker
 * [new branch] main -> main

```

The screenshot shows the Visual Studio Code editor with the `expense_tracker.py` file open. The file contains a Python script for an expense tracker. The terminal output shows the results of running the script, displaying a list of expenses and their total.

```
def view_by_category():
    print("\n60")
    for r in rows:
        print("{:4} {:10} {:12} {:8} {}".format(r["id"], r["date"], r["category"], r["amount"], r["note"]))

def menu():
    while True:
        print("\nPersonal Expense Tracker - Menu")
        print("(1) Add expense")
        print("(2) View all expenses")
        print("(3) View total spent")
        print("(4) View by category")
        print("(5) Exit")
        choice = input("Choose (1-5): ").strip()
        if choice == "1":
            add_expense()
        elif choice == "2":
            view_expenses()
        elif choice == "3":
            total_spent()
        elif choice == "4":
            view_by_category()
        elif choice == "5":
            print("Bye!")
            break
```

```
$ ls -la
total 41
drwxr-xr-x 1 agrim 107609  0 Nov 23 17:24 ./
drwxr-xr-x 1 agrim 107609  0 Nov 23 16:04 ../
drwxr-xr-x 1 agrim 107609  0 Nov 23 17:32 .git/
drwxr-xr-x 1 agrim 107609  0 Nov 23 16:13 docs/
-rwxr-xr-x 1 agrim 107609 4119 Nov 23 16:28 expense_tracker.py
-rw-r--r-- 1 agrim 107609 177 Nov 23 16:38 expenses.csv
drwxr-xr-x 1 agrim 107609  0 Nov 23 16:16 screenshots/
```

6). Git version checking

The screenshot shows the Visual Studio Code editor with the `expense_tracker.py` file open. The terminal output shows the result of running the `git --version` command, indicating that Git version 2.52.0 is installed on Windows.

```
PS C:\Users\agrim\OneDrive\Documents\expense-tracker> git --version
git version 2.52.0.windows.1
PS C:\Users\agrim\OneDrive\Documents\expense-tracker>
```